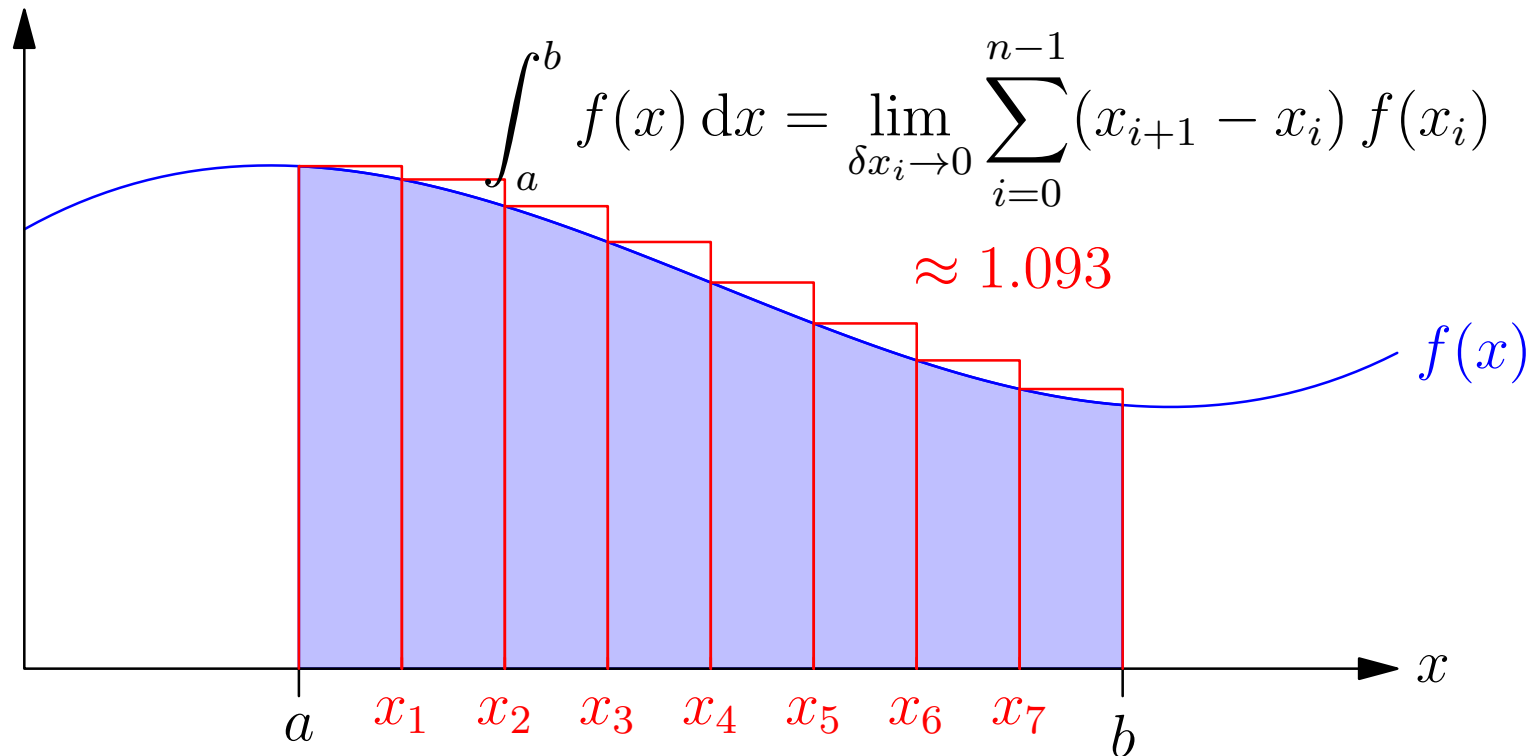


Advanced Machine Learning

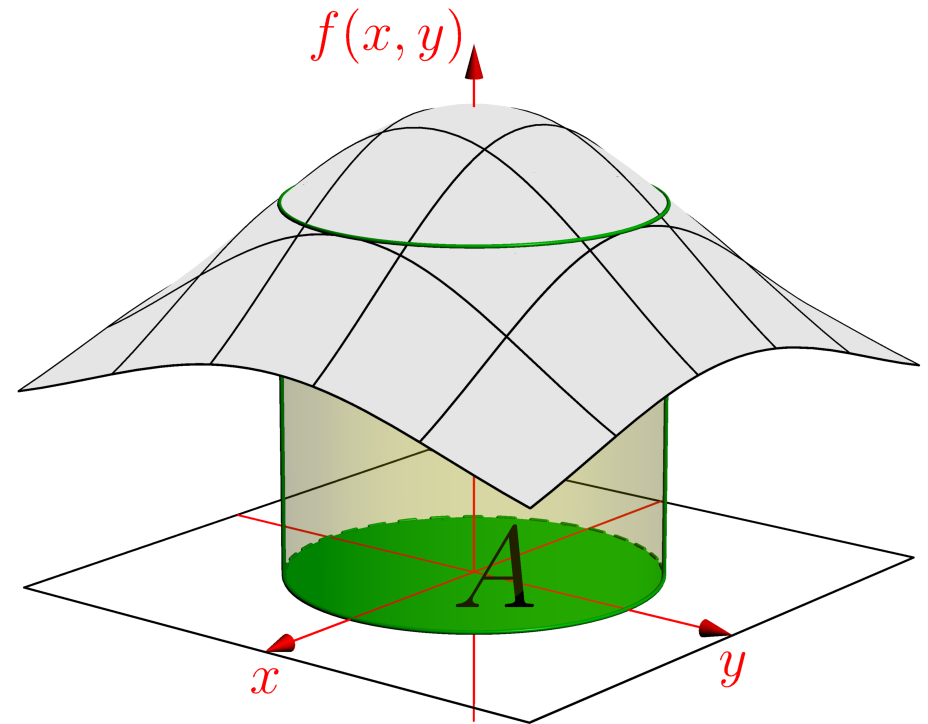
Integral Calculus



Riemann Integration, integration by parts, gaussian integrals

Outline

1. **Defining Integrals**
2. Doing Integrals
3. Gaussian Integrals

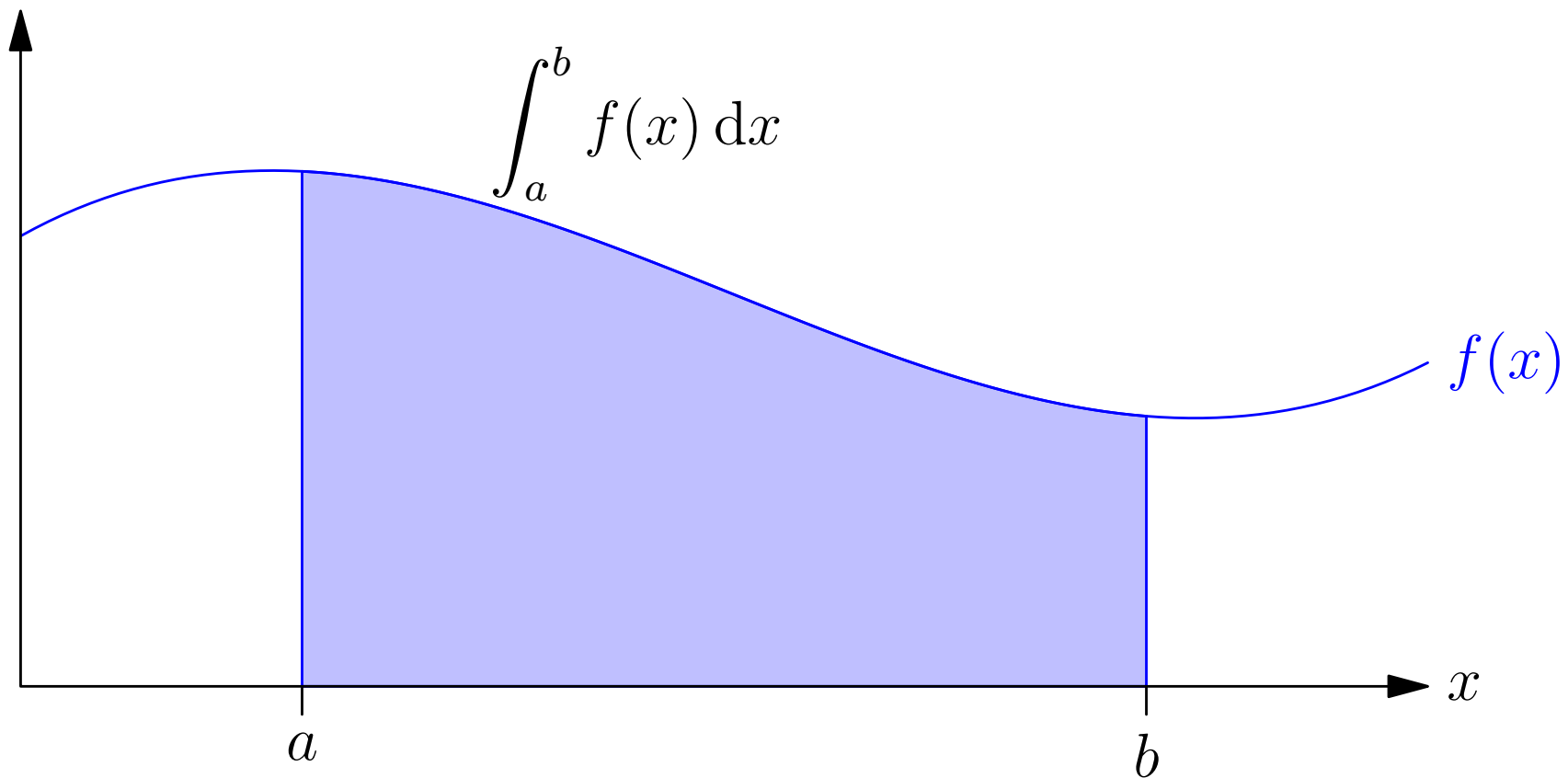


Riemann Integral

- Integrals represent area beneath a curve

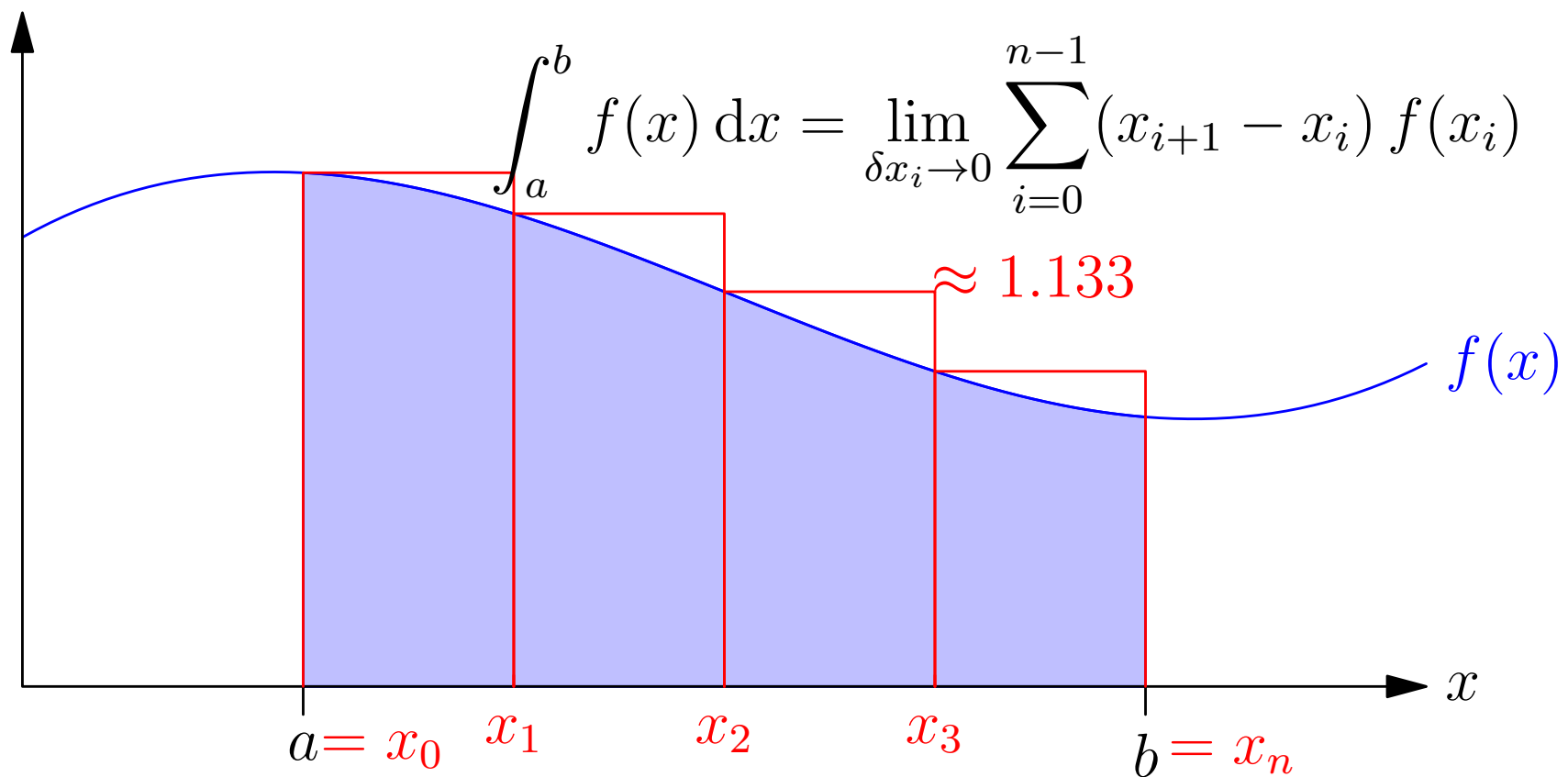
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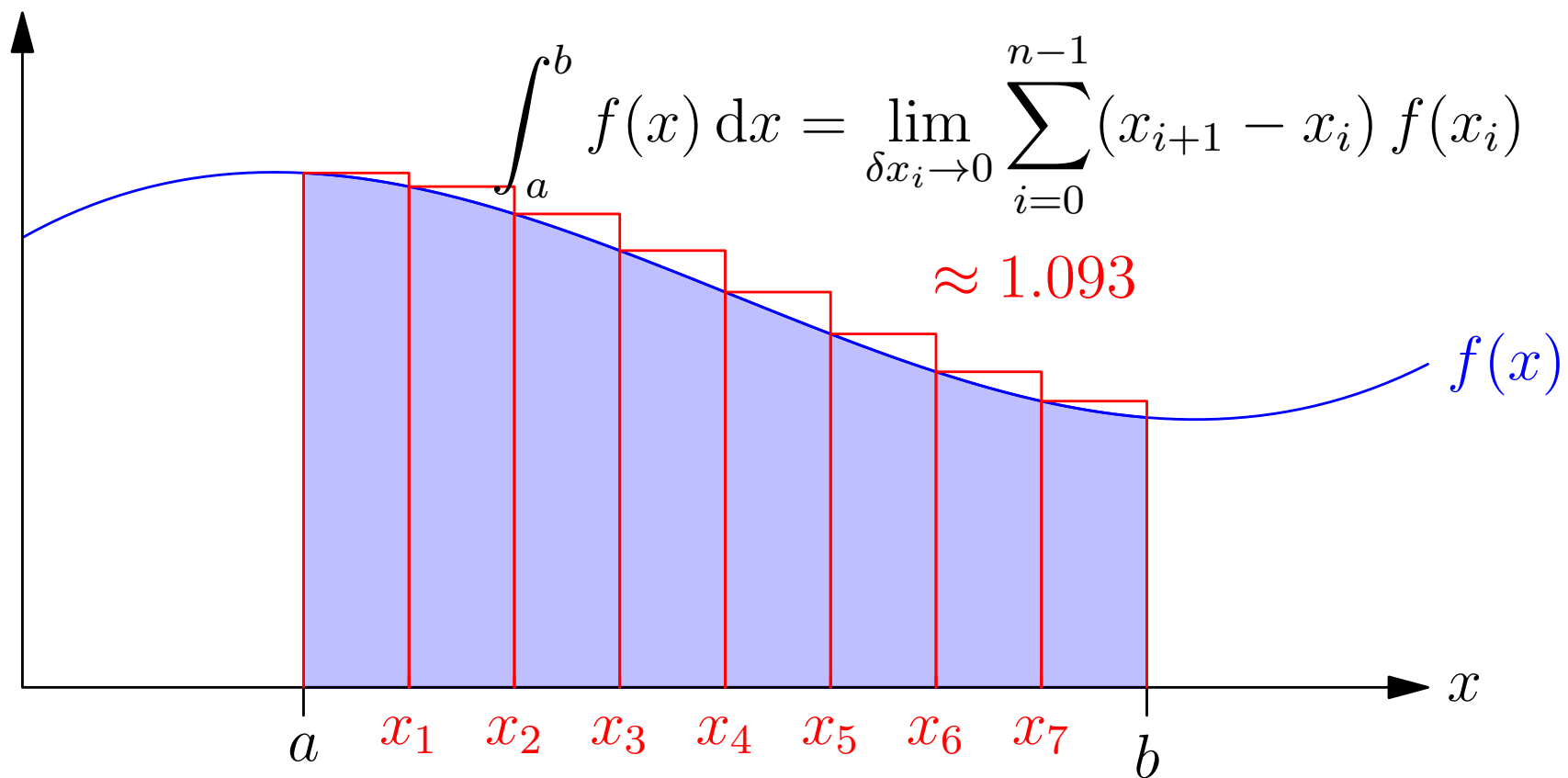
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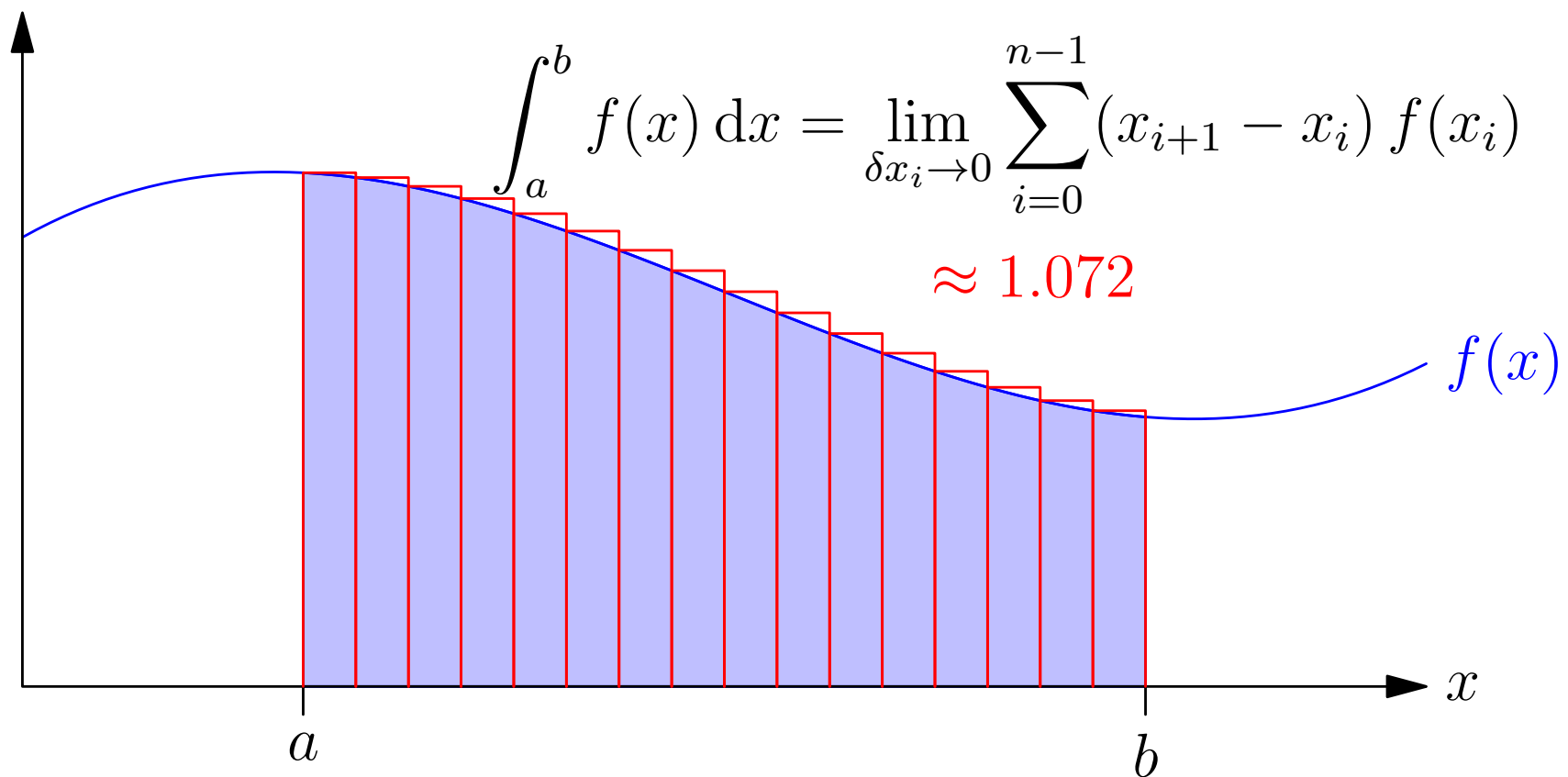
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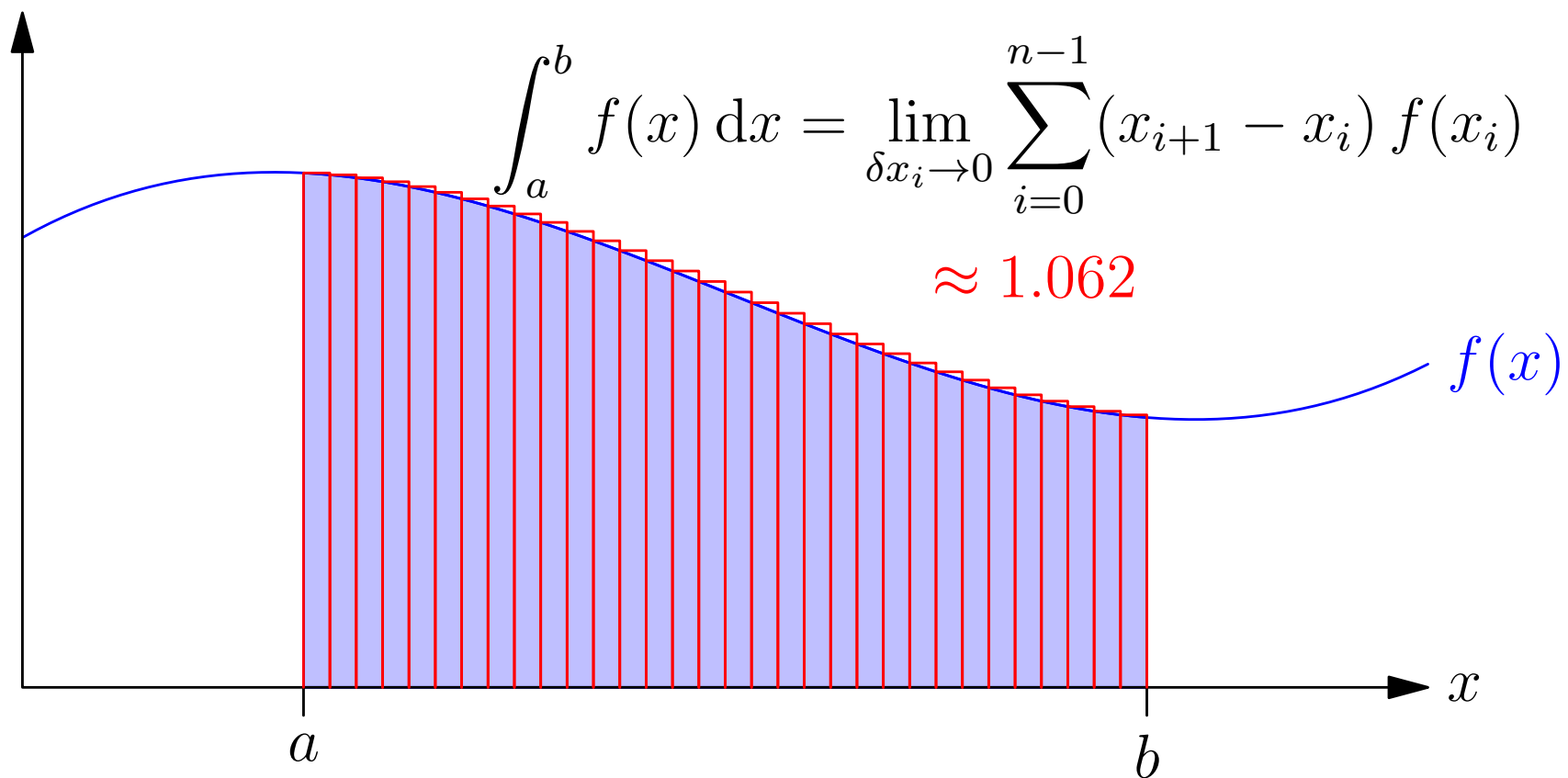
Riemann Integral

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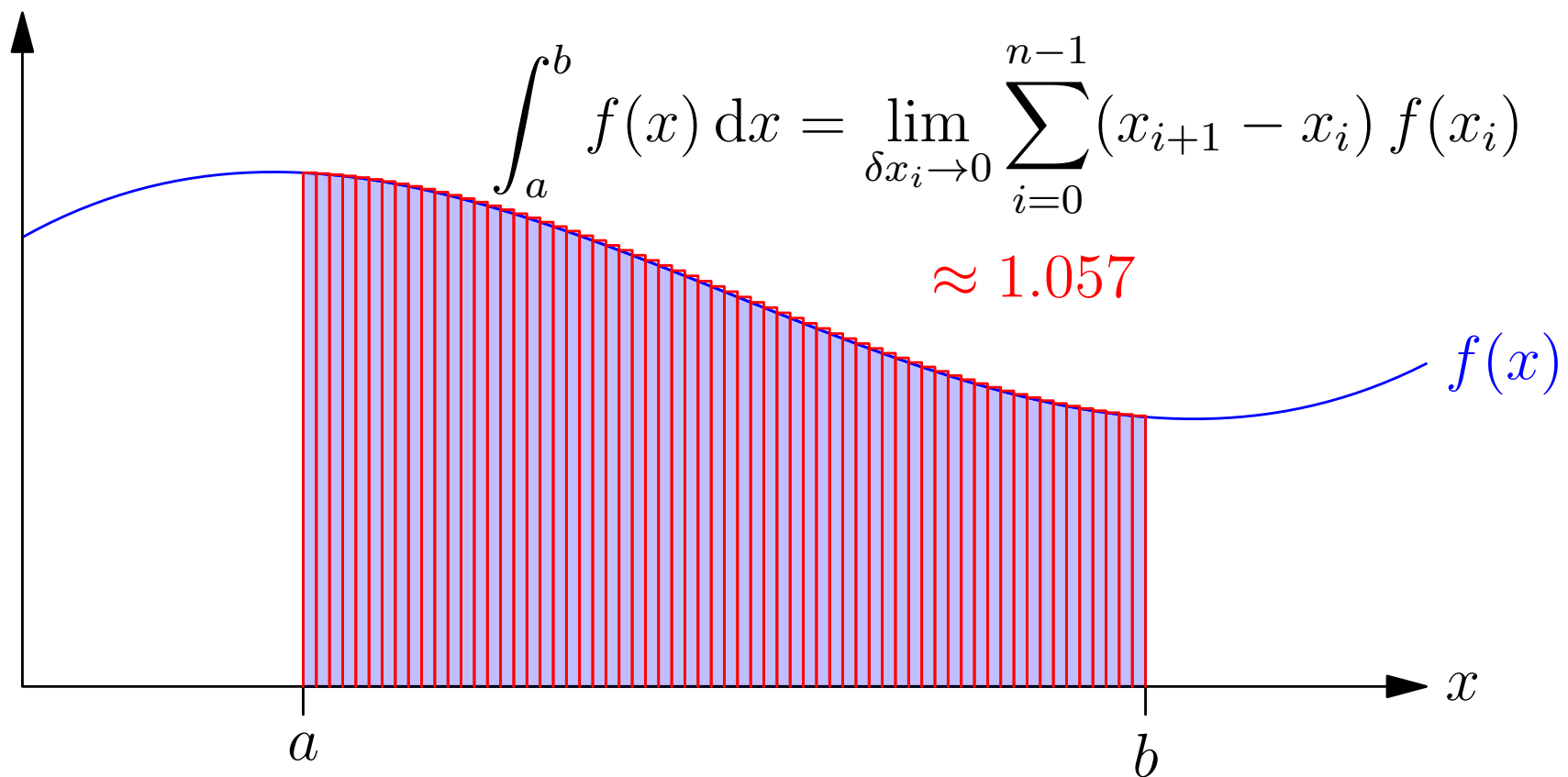
Riemann Integral

- Integrals represent area beneath a curve



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Linearity of Integration

- Integration is a linear operator

$$\int_a^b (r f(x) + s g(x)) dx = \lim_{\delta x_i \rightarrow 0} \sum_{i=0}^{n-1} (x_{i+1} - x_i) (r f(x_i) + s g(x_i))$$

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Fundamental Law of Calculus

- Let

$$I(a, x) = \int_a^x f(z) \mathrm{d}z = \lim_{\delta z_i \rightarrow 0} \sum_{i=0}^{n-1} (z_{i+1} - z_i) f(z_i)$$

- Now for small δx

$$I(a, x + \delta x) = \int_a^{x+\delta x} f(z) \mathrm{d}z = \lim_{\delta z_i \rightarrow 0} \sum_{i=0}^{n-1} (z_{i+1} - z_i) f(z_i) + \delta x f(x)$$

- Thus

$$\frac{\mathrm{d}I(a, x)}{\mathrm{d}x} = \lim_{\delta x \rightarrow 0} \frac{I(x + \delta x) - I(x)}{\delta x}$$

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The Other Way Around

- Consider

$$\int_a^b \frac{df(x)}{dx} dx = \int_a^b \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x} dx$$

The Other Way Around

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$$\begin{aligned}\int_a^b \frac{df(x)}{dx} dx &= \int_a^b \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x} dx \\ &= \lim_{x_{i+1} - x_i \rightarrow 0} \sum_{i=0}^{n-1} (x_{i+1} - x_i) \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}\end{aligned}$$

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- We can think of integration as an **anti-derivative** it undoes differentiation

Indefinite Integrals

- So far we have considered **definite integrals** where we integrate between two points (a and b)
- However, when think about integration as an anti-derivative, it is useful to think of a function $F(x) = \int f(x)dx$
- So that $F'(x) = f(x)$
- However the function $F(x)$, $F(x) + 1$, $F(x) + \pi$, etc. all have the same derivative so $F(x)$ is only defined up to an additive constant
- Note that the definite integral is given by

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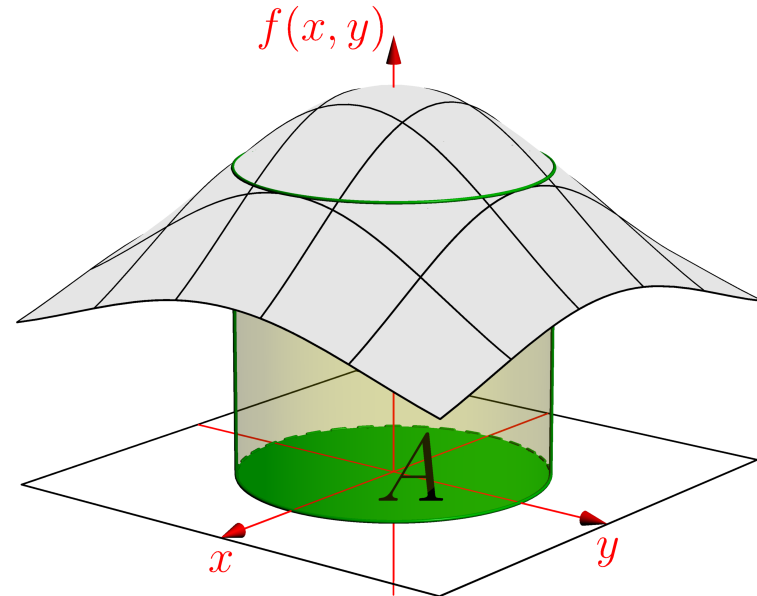
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Multiple Integrals

- For functions involving many independent variables (e.g. $f(x,y)$, $f(x,y,z)$, $f(\mathbf{x})$) we can integrate over multiple dimensions
- For example

$$\iint_A f(x,y) dx dy$$



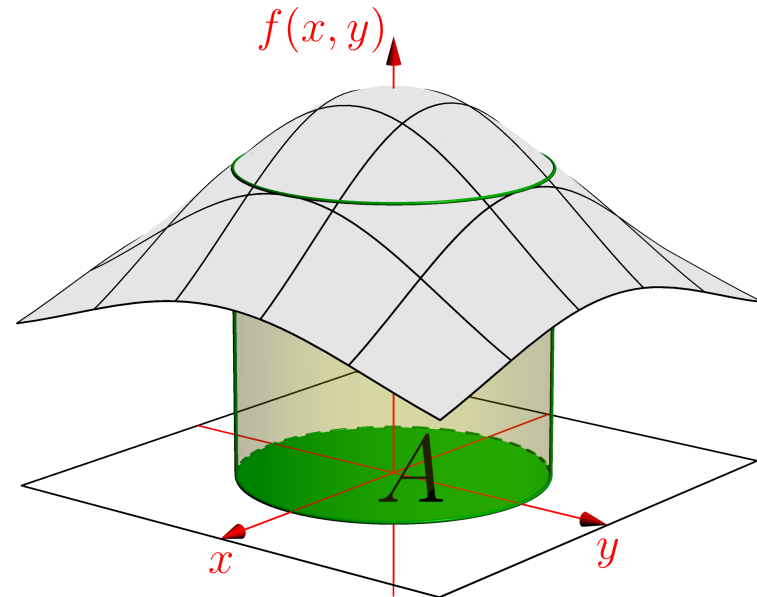
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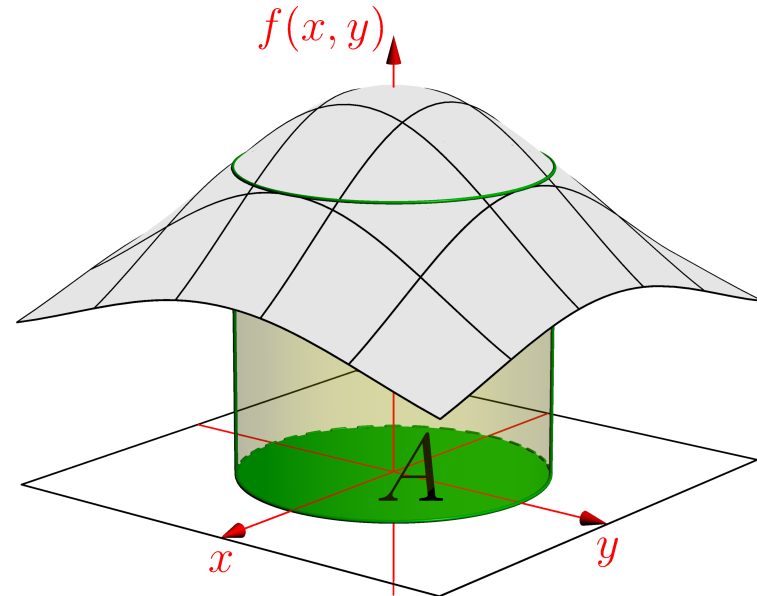
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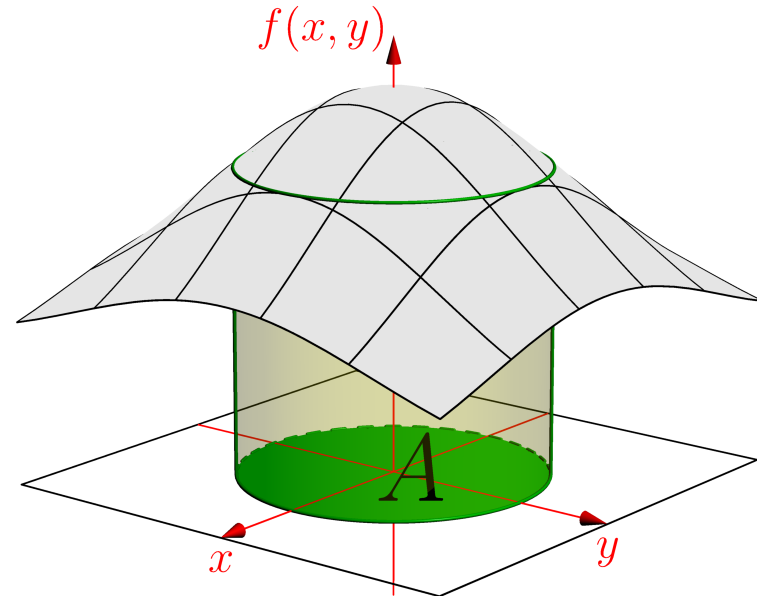
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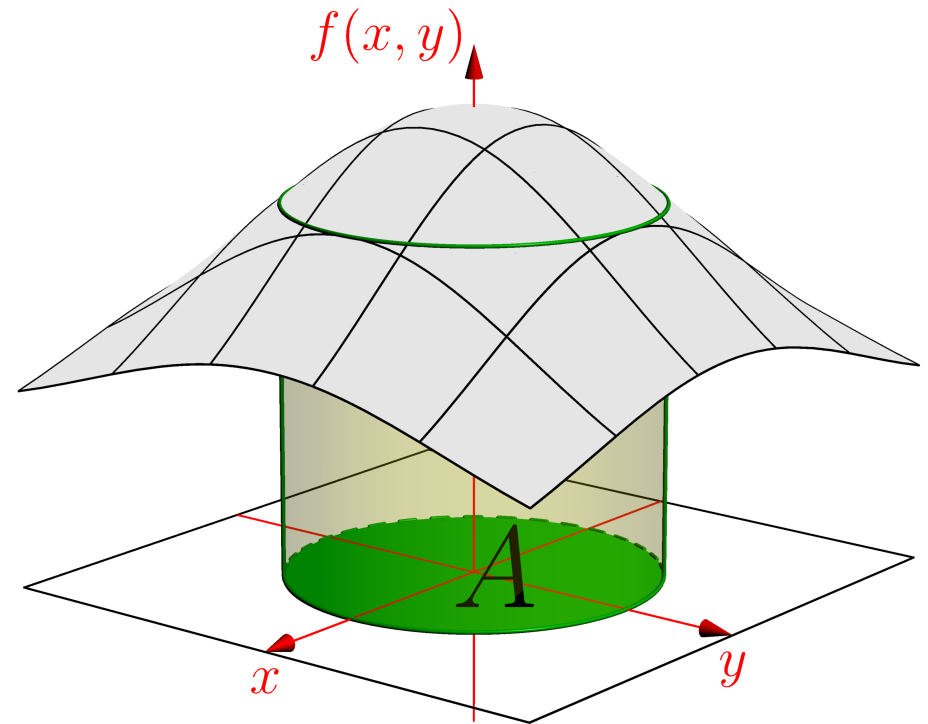


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$$\int \cdots \int f(x_1, x_2, \dots, x_n) dx_1 dx_2 \cdots dx_n = \int f(\mathbf{x}) d\mathbf{x}$$

Outline

1. Defining Integrals
2. **Doing Integrals**
3. Gaussian Integrals



Performing Integration

- A key method for performing integrals is through knowledge of the anti-derivative
- If we know $F'(x) = f(x)$ then $F(x) + c = \int f(x) dx$
- E.g. we know that $dx^n/dx = nx^{n-1}$ therefore

$$\int x^{n-1} dx = \frac{1}{n} \int \frac{dx^n}{dx} dx$$

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Is Integration Straightforward?

- We saw due to the product and chain rules that we can differentiate almost anything

- Products and compositions

$$\int f(x)g(x)dx = ?$$

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- Unfortunately, unlike differentiation we don't have a small parameter we can expand in
- In general integration is hard

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$$\begin{aligned}\int_a^b \frac{d}{dx}(f(x)g(x)) dx &= \int_a^b \frac{df(x)}{dx}g(x) dx + \int_a^b f(x)\frac{dg(x)}{dx} dx \\ &= [f(x)g(x)]_a^b\end{aligned}$$

- Unfortunately we get two integrals

Integration by Parts

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whether this is helpful depends on $f(x)$ and $g(x)$

Example of Integration by Parts

- Consider

$$\Pi(z) = \int_0^{\infty} x^z e^{-x} dx$$

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$$\begin{aligned}\Gamma(z) &= \int_0^\infty x^z e^{-x} dx = \int_0^\infty x^z \frac{d(-e^{-x})}{dx} dx \\ &= [x^z (-e^{-x})]_0^\infty - \int_0^\infty \frac{dx^z}{dx} (-e^{-x}) dx \\ &= \int_0^\infty (zx^{z-1}) e^{-x} dx = z \int_0^\infty x^{z-1} e^{-x} dx = z\Gamma(z-1)\end{aligned}$$

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$$\Pi(0) = \int_0^\infty e^{-z} dz = [-e^{-x}]_0^\infty$$

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- Now

$$\Pi(n) = n\Pi(n-1) = n(n-1)\Pi(n-2) = n(n-1)(n-2)\dots 1 = n!$$

Substitution

- We can make a transformation from x to u

$$\int_a^b f(x) dx = \lim_{\delta x_i \rightarrow 0} \sum_{i=0}^{n-1} f(x_i)(x_{i+1} - x_i)$$

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★ where u_i is such that $x(u_i) = x_i$ or $u_i = u(x_i)$ where $u(x)$ is the inverse of $x(u)$

★ using $\lim_{\delta u_i \rightarrow 0} \frac{x(u_{i+1}) - x(u_i)}{u_{i+1} - u_i} = \frac{dx(u)}{du}$

Example of Integration by Substitution

- We consider $I(n) = \int_0^{\infty} x^n e^{-x^2/2} dx$
- Let $u(x) = x^2/2$ or $x(u) = \sqrt{2u}$ so that

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Changing Variables in Multidimensional Space

- When changing variables in many dimensions $\mathbf{x} \rightarrow \mathbf{u}$ the change of variables involves the Jacobian

$$\int f(\mathbf{x}) d\mathbf{x} = \int f(\mathbf{x}(\mathbf{u})) |\det(\mathbf{J})| d\mathbf{u}, \quad \mathbf{J} = \begin{pmatrix} \frac{\partial x_1}{\partial u_1} & \frac{\partial x_1}{\partial u_2} & \cdots & \frac{\partial x_1}{\partial u_n} \\ \frac{\partial x_2}{\partial u_1} & \frac{\partial x_2}{\partial u_2} & \cdots & \frac{\partial x_2}{\partial u_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial u_1} & \frac{\partial x_n}{\partial u_2} & \cdots & \frac{\partial x_n}{\partial u_n} \end{pmatrix}$$

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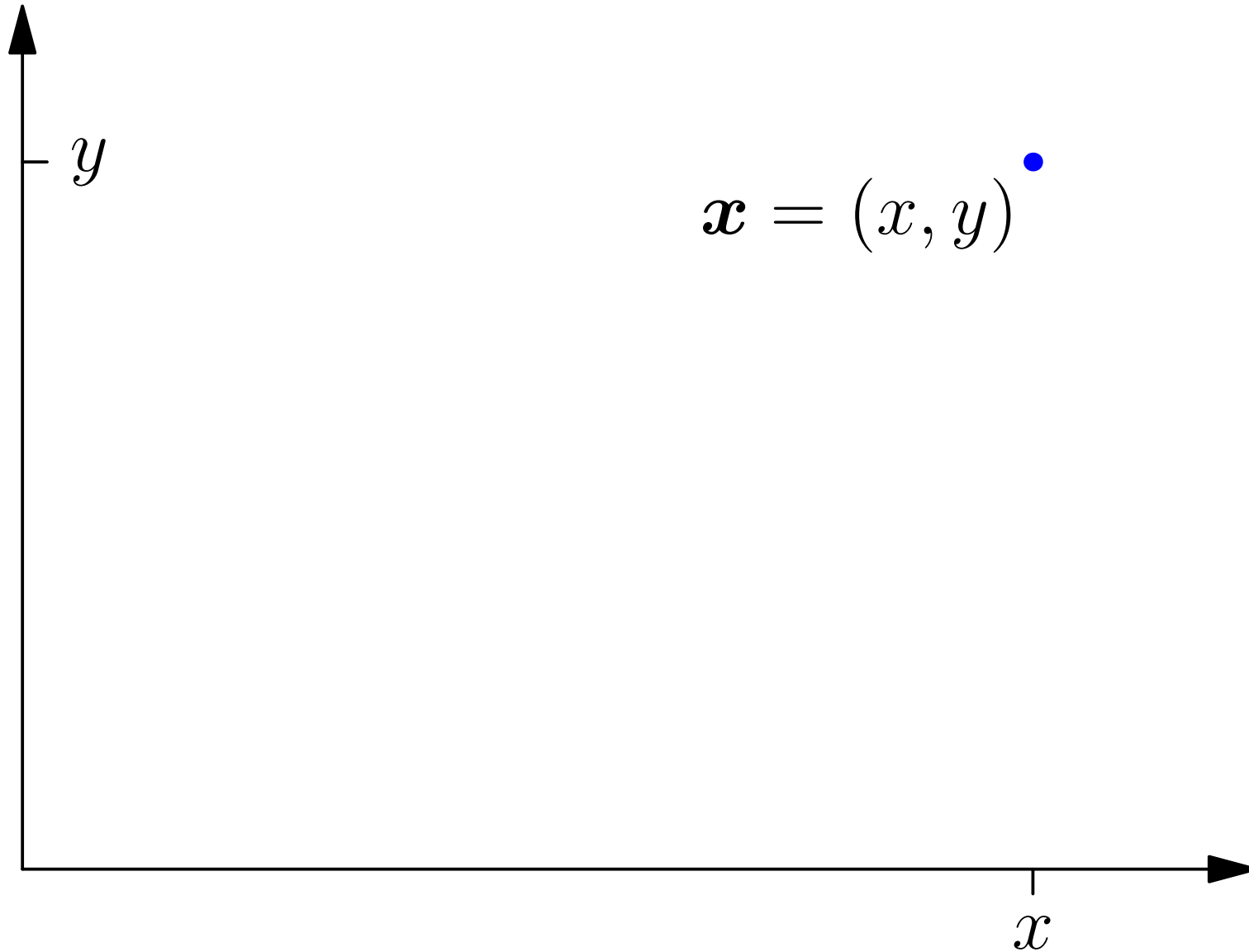
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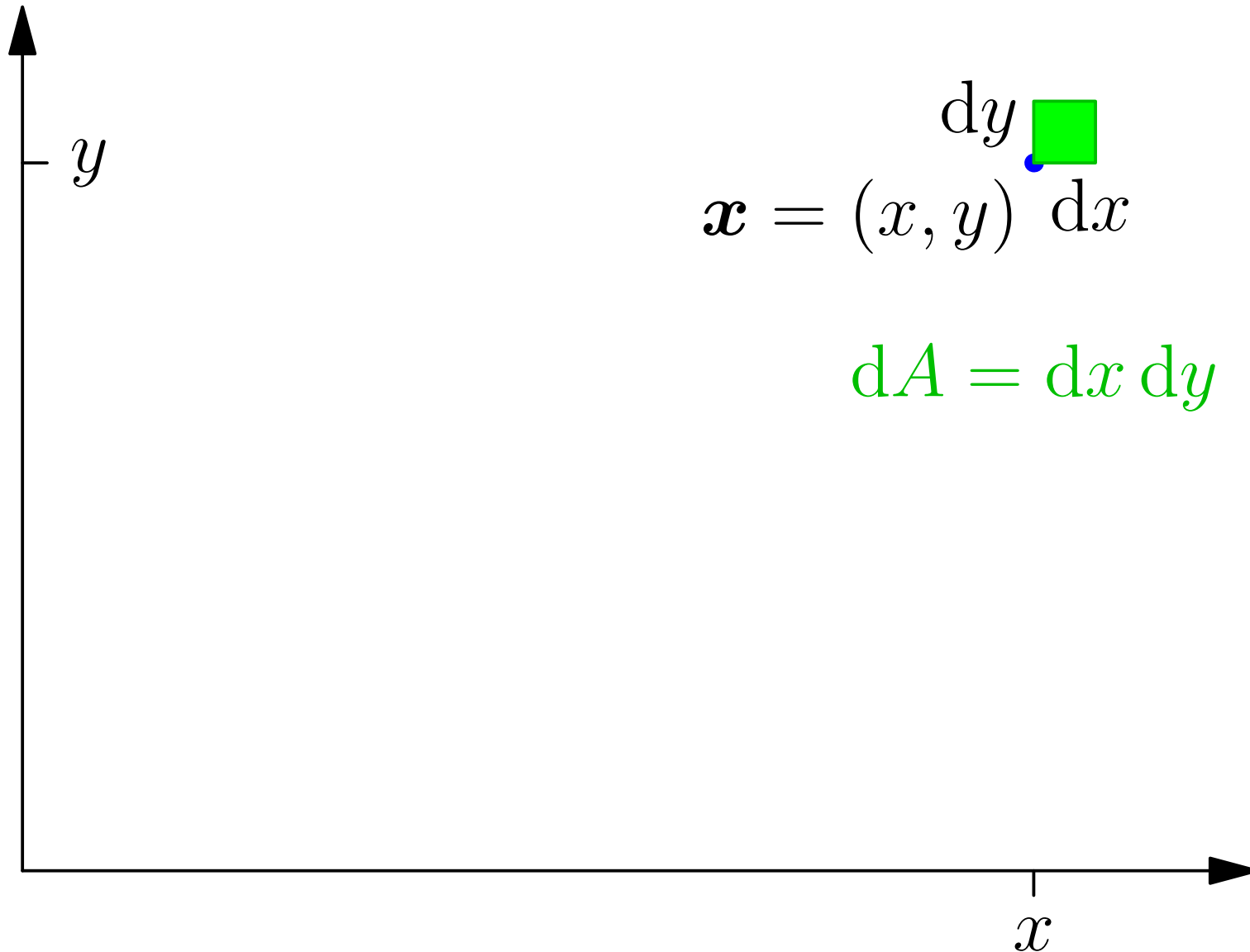
$$\begin{aligned} |\det(\mathbf{J})| &= \left| \det \begin{pmatrix} \frac{\partial r \cos(\theta)}{\partial r} & \frac{\partial r \cos(\theta)}{\partial \theta} \\ \frac{\partial r \sin(\theta)}{\partial r} & \frac{\partial r \sin(\theta)}{\partial \theta} \end{pmatrix} \right| = \left| \det \begin{pmatrix} \cos(\theta) & -r \sin(\theta) \\ \sin(\theta) & r \cos(\theta) \end{pmatrix} \right| \\ &= r (\cos^2(\theta) + \sin^2(\theta)) = r \end{aligned}$$

- That is, $dx dy = r dr d\theta$

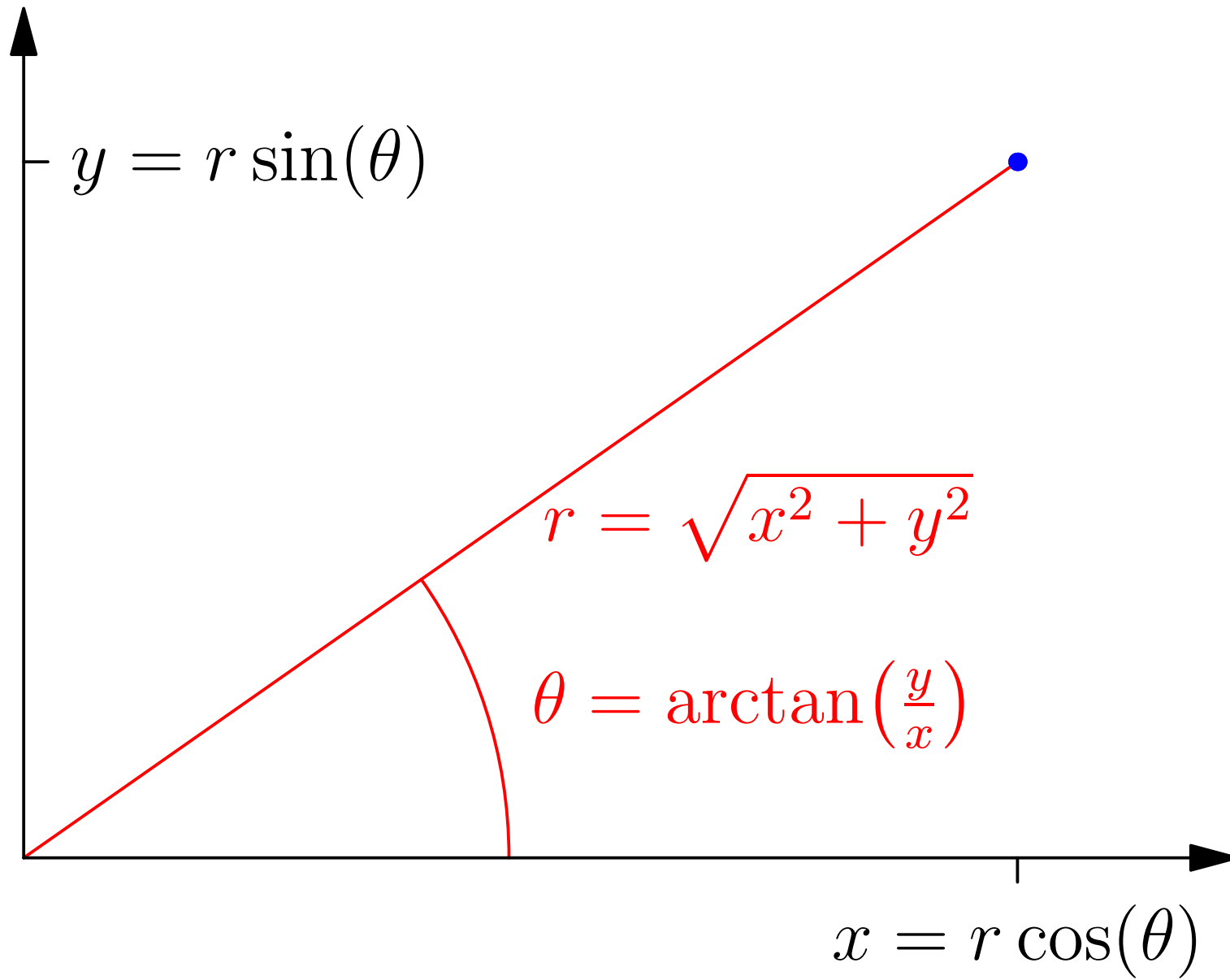
Change of Variables in Pictures



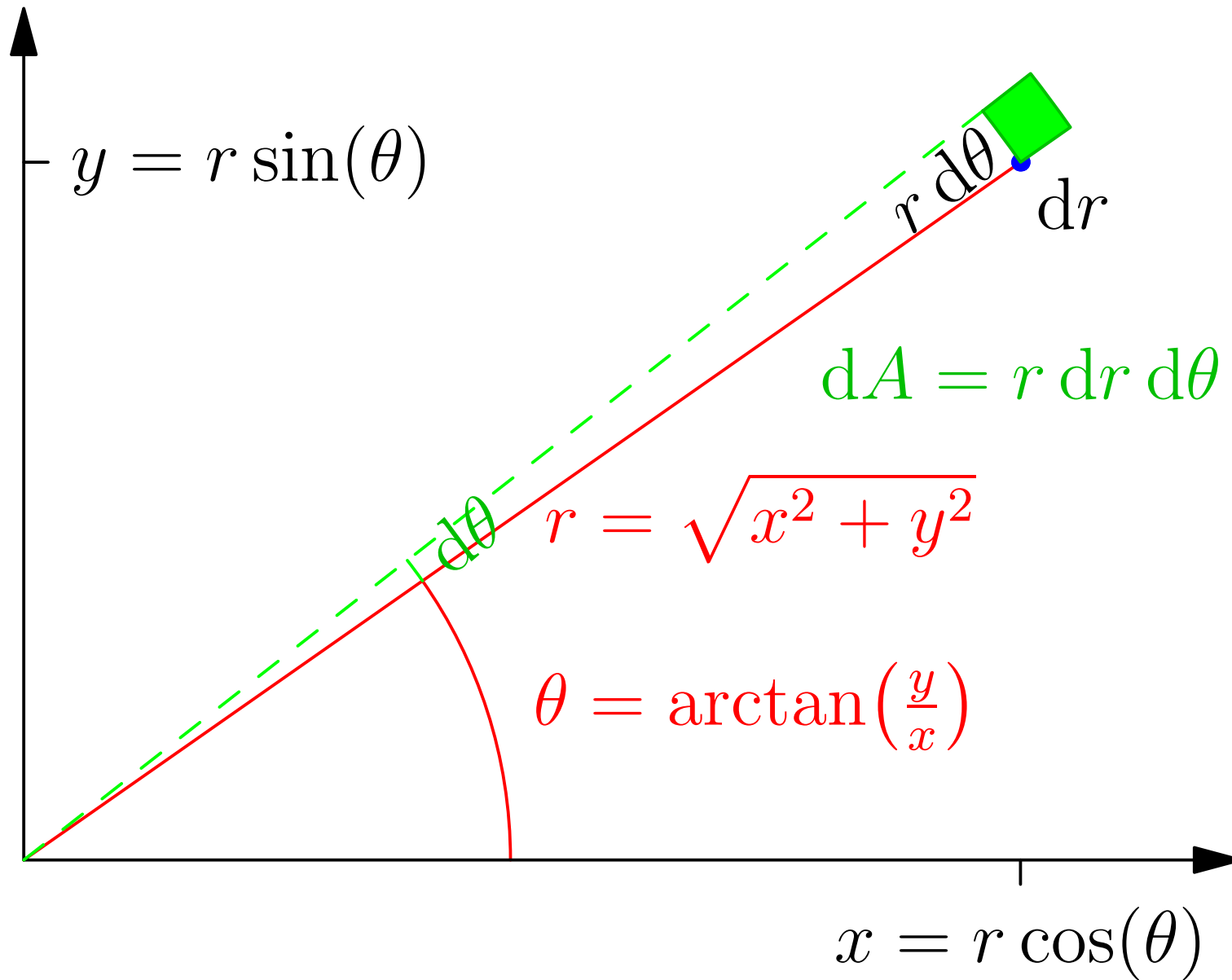
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Differentiating Through the Integral

- A trick that sometimes works is differentiating through an integral, e.g. consider finding moments

$$M_n = \mathbb{E}[X^n] = \int_{-\infty}^{\infty} x^n f_X(x) dx$$

- We can define a momentum generating function

$$Z(\ell) = \int_{-\infty}^{\infty} e^{\ell x} f_X(x) dx$$

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- Note that $e^{\ell x} = 1 + \ell x + \frac{1}{2}\ell^2 x^2 + \frac{1}{3!}\ell^3 x^3 + \dots$

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Special Functions

- There are integrals with no known closed form solution
- We saw that $\Pi(z) = \int_0^{\infty} x^z e^{-x} dx$ satisfies $\Pi(z) = z\Pi(z-1)$
- For integer n then $\Pi(n) = n!$, but for general z , the integral $\Pi(z)$ can't be written in terms of elementary functions
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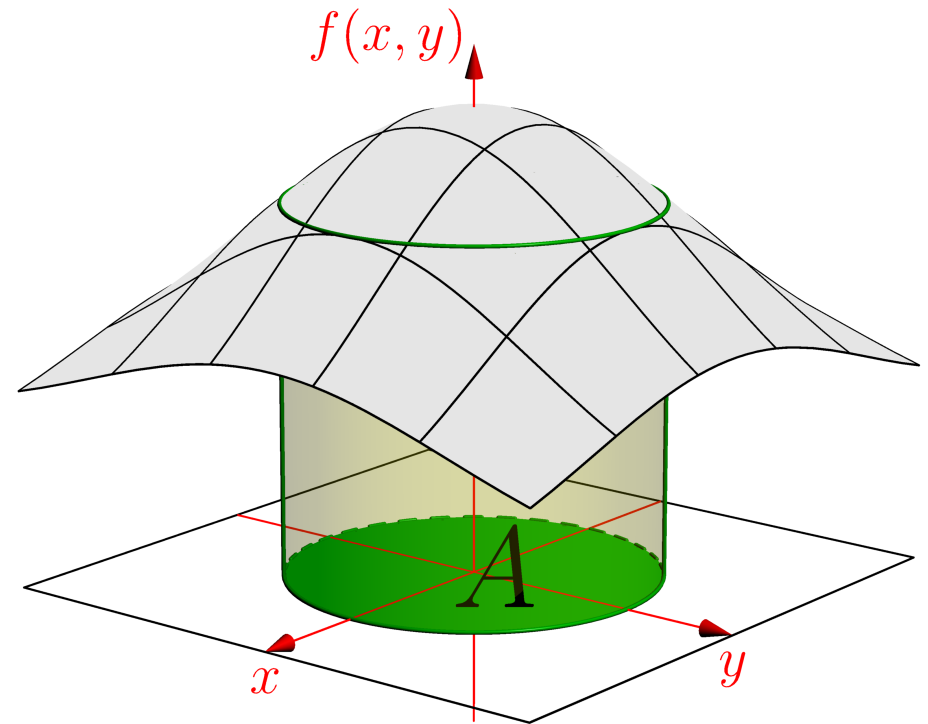
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- Other special function defined by integrals exist (e.g. the Bessel , Aire, hypergeometric, elliptic, error functions, . . .)

Outline

1. Defining Integrals
2. Doing Integrals
3. **Gaussian Integrals**



Gaussian Integrals

- Gaussian integrals are integrals involving e^{-x^2} , e.g.

$$\int_{-\infty}^{\infty} e^{-x^2} dx$$

$$\int_{-\infty}^{\infty} x^4 e^{-ax^2 - bx} dx$$

- They are important in computing integrals with respect to the normal distribution

$$\mathcal{N}(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}$$

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- The great news is that these integrals are all doable
- The bad news is that they are quite tricky to do

The Gaussian Integral

- The integral over a Gaussian is surprisingly difficult

$$I_1 = \int_{-\infty}^{\infty} e^{-x^2/2} dx$$

- There is a nice trick which is to consider

$$I_1^2 = \int_{-\infty}^{\infty} e^{-x^2/2} dx \int_{-\infty}^{\infty} e^{-y^2/2} dy = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)/2} dx dy$$

- Making the change of variables $r = \sqrt{x^2 + y^2}$ and $\theta = \arctan(y/x)$ (so that $x = r \cos(\theta)$, $y = r \sin(\theta)$ and $x^2 + y^2 = r^2$)

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- Finally let $u = r^2/2$ so that $du/dr = r$ or $du = r dr$ we get

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- Incidentally, $I_1 = \sqrt{2}\Pi(-1/2)$ so $\Pi(-1/2) = \Gamma(1/2) = \sqrt{\pi}$

Normal Distribution

- We consider

$$I_2 = \int_{-\infty}^{\infty} e^{-(x-\mu)^2/(2\sigma^2)} dx$$

- Making the change of variables $z = (x - \mu)/\sigma$ so that $dz = dx/\sigma$ or $dx = \sigma dz$. Then

$$I_2 = \sigma \int_{-\infty}^{\infty} e^{-z^2/2} dz = \sigma I_1 = \sqrt{2\pi} \sigma$$

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where $\mathbf{x} = (x_1, x_2, \dots, x_n)^\top$

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$$\begin{aligned} I_3 &= \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} e^{-\frac{1}{2} \sum_{i=1}^n x_i^2} dx_1 \cdots dx_n \\ &= \prod_{i=1}^n \int_{-\infty}^{\infty} e^{-x_i^2/2} dx_i = \prod_{i=1}^n \sqrt{2\pi} = (2\pi)^{n/2} \end{aligned}$$

Full Multi-variate Normal

- Consider

$$I_4 = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \boldsymbol{\Xi}^{-1}(\mathbf{x}-\boldsymbol{\mu})} d\mathbf{x}_1 \cdots d\mathbf{x}_n$$

- Let $\boldsymbol{\Xi}^{-1} = \mathbf{V}\boldsymbol{\Lambda}^{-1}\mathbf{V}^T$ and make the change of variables $\mathbf{y} = \mathbf{V}^T(\mathbf{x} - \boldsymbol{\mu})$
- The Jacobian $\mathbf{J}$ has elements (note that $\mathbf{x} = \mathbf{V}\mathbf{y} + \boldsymbol{\mu}$)

$$J_{ij} = \frac{\partial x_i}{\partial y_j} = \frac{\partial}{\partial y_j} \left(\sum_{k=1}^n V_{ik} y_k + \mu_i \right) = V_{ij}$$

- So that $\mathbf{J} = \mathbf{V}$ and consequently $|\det(\mathbf{J})| = |\det(\mathbf{V})| = 1$ then

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Determinants

- Using the facts, that $\Xi = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$ then

$$\det(\Xi) = \det(\mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top)$$

- Recall $I_4 = \prod_i \sqrt{2\pi\lambda_i}$
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- It occurs when you work with probabilities densities for continuous random variables
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- Integration is beautiful, but hard—often impossible
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- Making friends with integration will give you a super-power that not too many people share